

Indian Queen East

Lots 53, 54, 55 and 56

Owner's decision package

Development feasibility, disposition and compensation options

9588, 9584, 9580 and 9576 Fort Foote Road, Fort Washington

Prince George's County, Maryland · Zone RSF-95 · Plat Book WWW 65, folio 60

Prepared 2026-09-11

The finding in one paragraph

All four lots lie in the floodplain of North Branch Broad Creek, held back by the Fort Foote Road embankment. Lots 53 and 56 can carry houses. Lots 54 and 55 cannot: Lot 54 is 89 per cent under water in a storm that has a one-in-two chance of happening this year, and no engineering measure available to you changes that. The recommendation is to build two houses and place the other two lots under a floodplain easement — which is also what makes the two houses permissible.

What this document is, and is not

It summarises a preliminary engineering study for the purpose of a commercial decision. It is not a sealed engineering document, not an appraisal, not a legal opinion and not tax advice. Values, deductions and conveyancing all require your own professionals. Everything here should be confirmed by a Maryland Professional Engineer before it is relied on or submitted.

1. Where the four lots stand

The lots sit at the bottom of a 397-acre watershed, immediately upstream of the point where North Branch Broad Creek passes under Fort Foote Road. The road embankment acts as a dam. Water backs up behind it and stands across the rear of all four properties.

The 100-year water surface is EL 55.47. The road itself sags to EL 54.00 and is overtopped in storms more frequent than one in ten years.

How often each lot is under water

Storm	Chance in any year	Lot 53	Lot 54	Lot 55	Lot 56
2-year	50%	24%	89%	41%	0%
5-year	20%	63%	97%	78%	2%
10-year	10%	77%	99%	92%	11%
25-year	4%	81%	100%	93%	14%
50-year	2%	84%	100%	95%	16%
100-year	1%	88%	100%	95%	20%

Percentage of each lot standing under water at the peak of the storm, existing ground.

Lot 54 is under water more often than not

A 2-year storm is not a rare event. It has a 50 per cent chance of occurring in any given year, and in it Lot 54 is 89 per cent submerged and Lot 55 is 41 per cent. Over a thirty-year mortgage these are near-certainties, repeatedly.

Why it cannot be engineered away

Three routes were tested and each was quantified.

- **Enlarging the road culvert.** Even a bridge-class structure leaves the 100-year surface at EL 48.77, still several feet above the rear of Lots 54 and 55. It would also push up to 80 per cent more flow onto the properties downstream, which the county will not accept. It is a county roadway project in any case, not something you can do.
- **Excavating storage.** Holding the water down to EL 50 would take 136,670 cubic yards of excavation — about 49 feet deep across all four properties. It would also not work: the rear of Lots 54 and 55 is already at or below the water level in the creek downstream, so anything dug there fills and stays full.
- **Diverting upstream drainage.** Removing the entire Oxon Hill Middle School property from the watershed — 21 acres, 5 per cent of the catchment — lowers the 100-year water surface by 0.07 feet. Under an inch.

2. What can be built

Lots 53 and 56 carry houses. Their building envelopes stand on the higher ground toward Fort Foote Road, and with the design in the study both dwellings finish clear of the floodplain in plan as well as above it in elevation.

Lot	Before grading	After grading	Finished floor	Above the water
Lot 53	51 ft inside the floodplain	18 ft clear of it	EL 59.24	+3.72 ft
Lot 56	10 ft inside the floodplain	29 ft clear of it	EL 61.02	+5.50 ft

Distance from the nearest corner of the dwelling to the 100-year floodplain limit.

Neither house may have a basement, and both are designed on vented crawlspace foundations with flood openings, so the structure is lifted rather than the ground filled. The rear of both lots stays in the floodplain and would be covered by the easement. Flood insurance will be required whatever the floor elevation, and a lender will insist on it.

Why dedicating Lots 54 and 55 is what makes the other two work

Any fill placed below the flood level displaces water and must be replaced by digging an equal volume out somewhere else. That is the rule that was blocking the four-house scheme:

	Volume
Fill below the flood level, four houses	5,087 cy
Of which comes from Lots 54 and 55	4,196 cy (83%)
Fill below the flood level, Lots 53 and 56 only	891 cy
Excavation available on Lots 54 and 55	1,399 cy
Ratio available to required	1.6 : 1

The two lots that cannot be built on are the two lots that supply what the buildable pair needs. Dedicating them is not writing them off; it is spending them on the approval.

A margin, not a cushion

The ratio is 1.6 to 1 at an excavation floor that drains. It works, and it is not generous. A field survey may move it either way, and the design should not be committed to until the survey is in.

3. Routes to sell or be compensated for Lots 54 and 55

These are set out in the order a practitioner would normally test them. Several can run at the same time. None of the values below is an appraisal; obtaining one is the first step in every route.

A. Dedicate to the county as part of your own approval

The simplest route, and the one the engineering already supports. The floodplain easement is dedicated over Lots 54 and 55 by record plat or by deed, the compensatory storage is built within it, and the dedication is what carries the site plan for Lots 53 and 56 through review. Prince George's County has done exactly this within this subdivision before: Plat 118-083, recorded 17 January 1984, dedicated a 5.32-acre floodplain easement on the downstream side of Fort Foote Road.

You are not paid cash for this. You are paid in approval: the two houses become permissible. If the two buildable lots carry the project economics, this is usually the best-value route, and it is the fastest.

B. Sell the fee to Prince George's County

- **DPIE / Department of the Environment — drainage and floodplain acquisition.** The county acquires land for stormwater and floodplain purposes. Approach DPIE at pre-application with the study and ask directly whether the parcels are of interest. The argument is strong here: the land is already functioning as the county's flood storage behind a county road that overtops.
- **M-NCPPC — parkland acquisition.** The Maryland-National Capital Park and Planning Commission acquires open space, stream valley land and green infrastructure. This corridor is mapped in the county Green Infrastructure Plan. M-NCPPC has an acquisition programme and a budget cycle; enquiries go to the Department of Parks and Recreation land acquisition staff.
- **Maryland Program Open Space.** State funding administered through DNR that local jurisdictions use to buy land for conservation and recreation. It is applied for by the county or M-NCPPC, not by you, so in practice this route runs through route B above — but it is worth naming, because it is often what actually pays.

C. Conservation conveyance, with a tax deduction

- **Bargain sale to a land trust or public agency.** You sell below appraised fair market value and the difference is treated as a charitable gift. You receive cash and a deduction. A qualified appraisal is mandatory.
- **Donation in fee.** The whole value becomes a charitable deduction, subject to the federal limits on conservation gifts and to carry-forward rules. No cash.
- **Conservation easement, retaining ownership.** You keep title and sell or donate the development rights. The deduction is the difference in appraised value before and after. The Maryland Environmental Trust and local land trusts hold these. You remain liable for maintenance and, unless the assessment is also reduced, for the taxes.

Get the appraisal before you choose

Every route in section C turns on a qualified appraisal of the parcels as they actually are — floodplain land with no development potential. The appraised figure decides which route is worth pursuing, and the IRS requires one for any deduction above modest thresholds. This is the single cheapest thing you can do next.

D. Reduce what you are paying to hold them

Appeal the assessment. Lots 54 and 55 are almost certainly assessed as buildable residential lots. On this analysis they are not buildable. An appeal to the Maryland State Department of Assessments and Taxation, supported by the engineering study, should reduce the assessed value materially. This is money back every year, it does not depend on anyone else agreeing to buy anything, and it is independent of every other route on this list.

Appeals are filed with the local SDAT assessment office. There is a defined appeal window following each assessment notice, and an out-of-cycle petition route between notices. Your attorney or a property tax agent will know the current dates.

E. Resubdivision and consolidation

Rather than holding four lots, combine Lots 54 and 55 into Lots 53 and 56 as floodplain rear yard, or into a single outlot dedicated to drainage. This reduces the number of taxable parcels, removes two lots that cannot be sold as building lots, and can make the easement simpler to describe and record. It requires M-NCPPC approval of a resubdivision plat.

F. Sale to adjoining owners

Floodplain land has real value to a neighbour as rear-yard buffer, and none to a builder. The adjoining owners on either side are the most likely private buyers at any price. This is worth a letter; it is not worth a marketing campaign.

G. Environmental credit markets

Stream and wetland mitigation banking, and nutrient credit trading, both operate in Maryland. A restored and permanently protected stream corridor can generate credits that are sold to parties who need offsets. It requires a restoration design, an approved banking instrument and long-term stewardship, so it is slow and it has up-front cost. It is listed because the corridor here is a genuine candidate, not because it is easy.

H. Routes that will not work here

- **FEMA hazard mitigation buyouts.** These programmes target repetitively flooded structures. Vacant land does not qualify.
- **A FEMA map amendment (LOMA).** The lots are already outside FEMA's mapped flood zone. Being outside FEMA's map does not make them outside the county floodplain, and there is nothing for FEMA to amend.
- **Filling the lots out of the floodplain.** It would take 12,922 cubic yards below the flood level with nowhere on site to compensate it, and it would raise the water on the neighbours.

4. What to do next, in order

#	Action	Why now	Who
1	Field survey: datum tie to benchmarks H31A, H32A, H32B and the culvert inverts	Every number in the study rests on these two unmeasured items. One day of work.	Meekins Surveyors
2	Appraisal of Lots 54 and 55 as floodplain land	Decides which disposition route is worth pursuing, and is required for any deduction.	MAI appraiser
3	Assessment appeal on Lots 54 and 55	Independent of everything else. Recovers money annually.	Attorney or tax agent
4	DPIE pre-application meeting with the study	Ask whether the county wants the parcels, and put the road overtopping on the record.	You and your engineer
5	Obtain FPS 200546 from DPIE	The controlling floodplain study of record. Not in the file. Everything must reconcile to it.	Engineer
6	Maryland PE to review, correct and seal the study	Nothing is reviewable unsealed. The platform cannot supply this.	Water resources PE

Step 4 is also a duty

Fort Foote Road is overtopped in storms more frequent than one in ten years, carrying 215 cubic feet per second across the pavement and cutting emergency access. That condition exists whether or not you build. Reporting it to DPIE in writing costs nothing and protects you.

5. Limitations

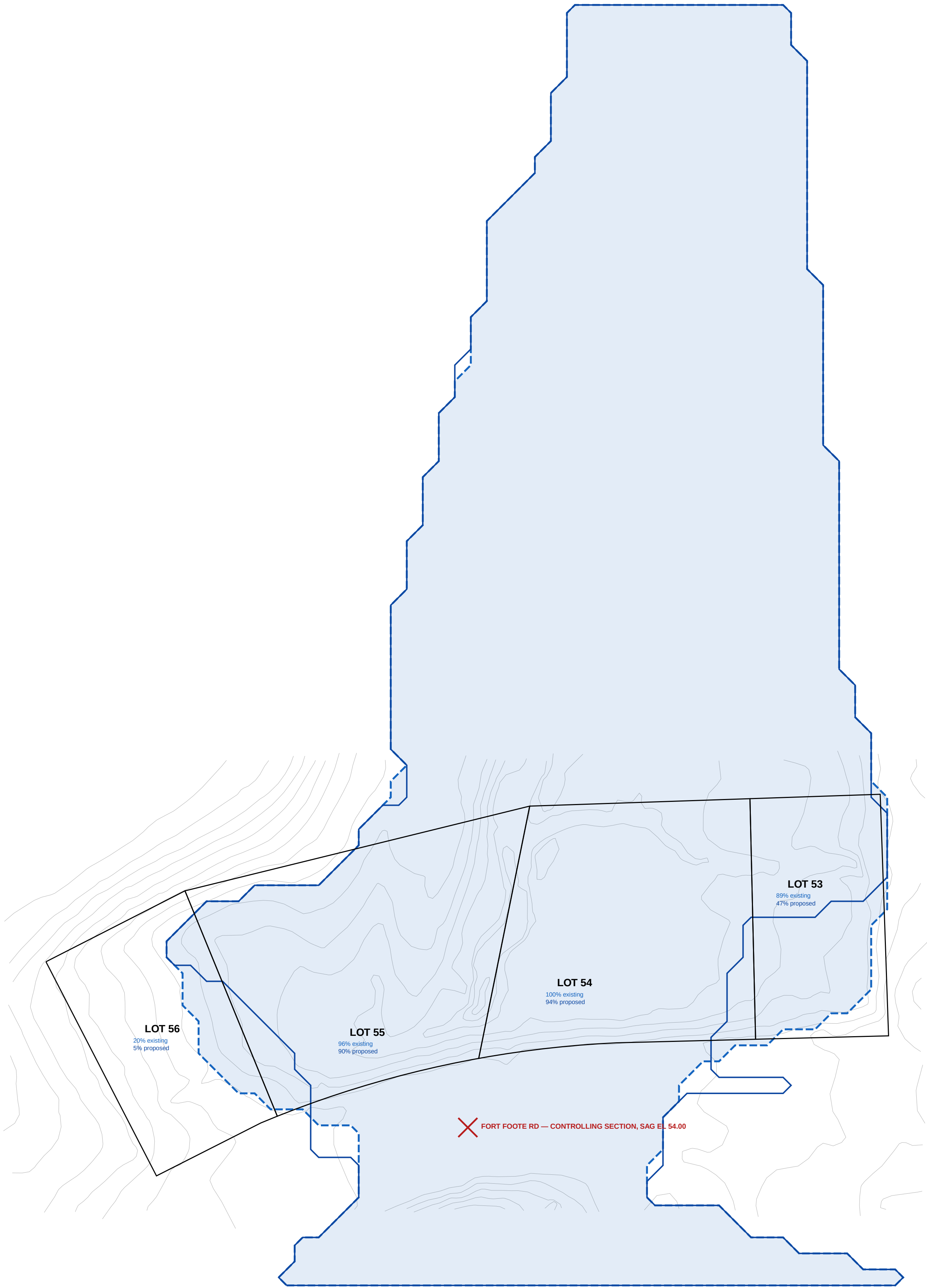
This package summarises a preliminary feasibility study. The terrain is published county LiDAR rather than a field survey. The existing culvert has never been measured and its size, inverts and entrance are assumed. HEC-RAS, the software of record, was not used; the same equations were solved directly. FPS 200546, the controlling county study, has not been obtained. Upstream stormwater facilities were not inventoried. No professional engineer has reviewed or sealed any part of this work.

The conclusions are robust to those gaps — Lot 54 is not 89 per cent flooded because of a survey error — but no part of this may be submitted as a sealed floodplain study, relied on in a conveyance, or used to support a tax position without professional review.

Accompanying documents: the floodplain and hydraulic study; sheets FP-101 delineation and impact, FP-102 mitigation and compensatory storage design, FP-103 floodplain delineation; and the GIS delineation files.

100-YEAR FLOODPLAIN DELINEATION — EXISTING AND PROPOSED CONDITIONS

North Branch Broad Creek at Fort Foote Road



0 100 FEET 1 in = 43 ft

KEALEE

Design, build, deliver

PROJECT

Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS

9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION

Prince George's County, Maryland

SHEET

FP-103 — FLOODPLAIN DELINEATION

DISCIPLINE

Maryland Professional Engineer

HORIZONTAL DATUM

NAD 83, Maryland State Plane, US survey feet (EPSG:2248)

VERTICAL DATUM

NAVD 88

WATERCOURSE

North Branch Broad Creek

CONTROLLING STRUCTURE

Fort Foote Road crossing, sag EL 54.00

DATE

2026-09-11

STATUS

PRELIMINARY. This delineation is derived from published county LIDAR terrain, not from a field survey, and is not a sealed floodplain study.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. _____ Date _____

WATER-SURFACE ELEVATIONS

EVENT	EXISTING	PROPOSED	AREA
10-year	EL 54.61	EL 54.62	3.95 ac
25-year	EL 54.93	EL 54.94	4.05 ac
50-year	EL 55.20	EL 55.22	4.16 ac
100-year	EL 55.51	EL 55.52	4.27 ac

AREA BELOW THE 100-YEAR SURFACE

LOT	LOT AREA	EXISTING	PROPOSED
LOT 53	12,500 sf	11,100 sf (89%)	5,900 sf (47%)
LOT 54	23,500 sf	23,500 sf (100%)	22,100 sf (94%)
LOT 55	25,600 sf	24,500 sf (96%)	23,000 sf (90%)
LOT 56	13,900 sf	2,800 sf (20%)	700 sf (5%)

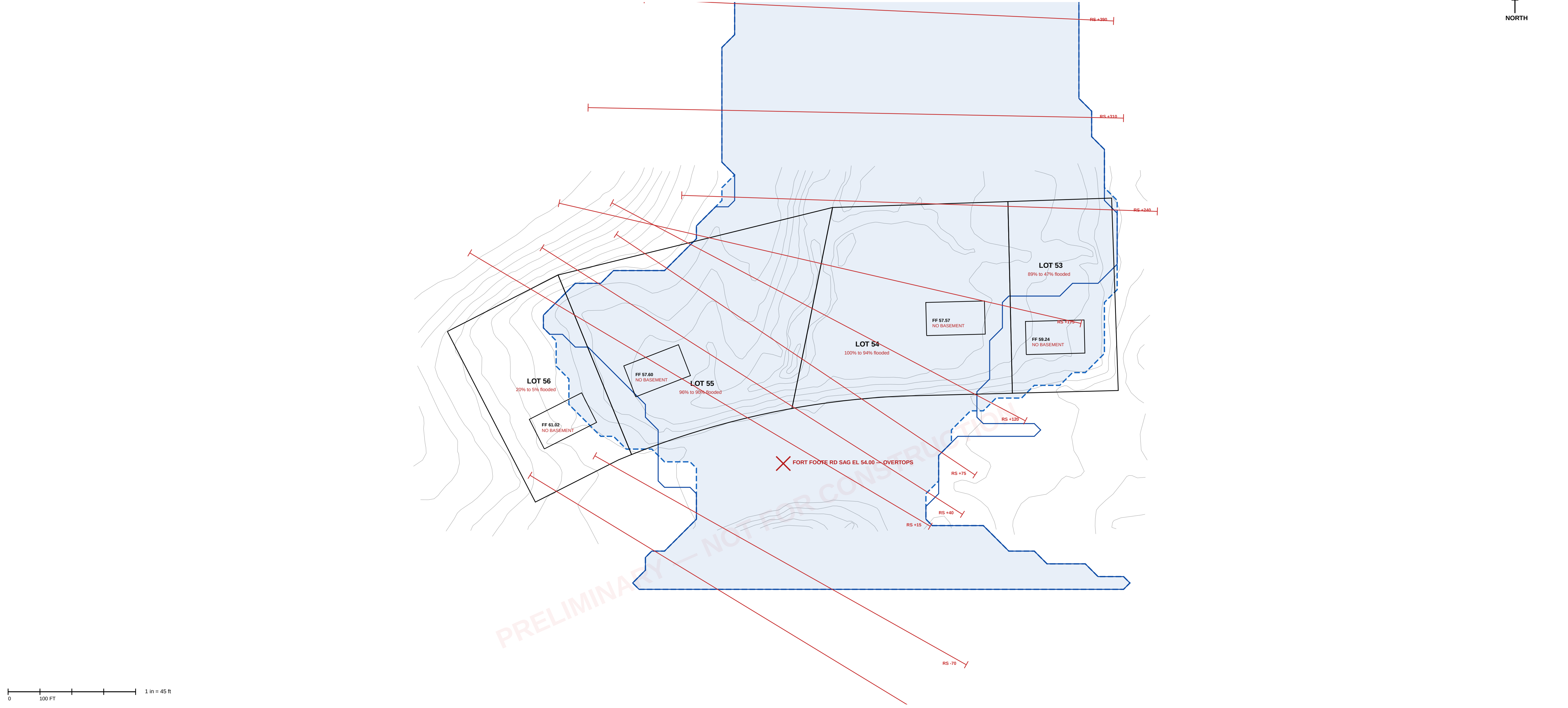
LEGEND

- 100-year floodplain limit — EXISTING condition
- 100-year floodplain limit — PROPOSED condition
- Lot boundary of record, Plat Book WWW 65 L 60
- Existing contour, 2 ft, county LIDAR (2023)
- Area below the 100-year water surface

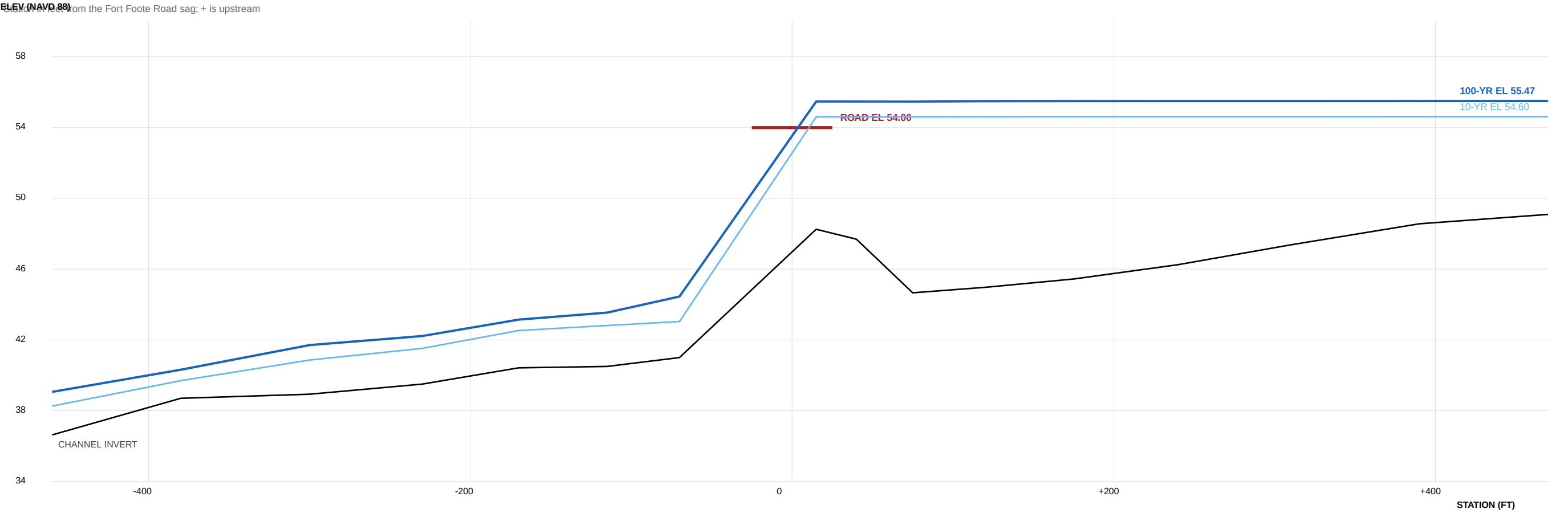
BASIS OF DELINEATION

The limits shown are the intersection of the modelled water surface with the terrain, flood-filled from the watercourse so that depressions not connected to it are excluded. Water surface elevations are from a one-dimensional steady standard-step model of North Branch Broad Creek through the Fort Foote Road crossing, with the crossing analysed by FPAWA HDS-5 and the roadway treated as a broad-crested weir. Discharges are by NRCS TR-55 on a 397-acre contributing area. Terrain is PGADAS Contour 2 Ft (2023), NAVD 88, NOT A FIELD SURVEY. The existing culvert has not been measured. FPS 200546, the controlling county study of record, has not been obtained. See the floodplain study for the full basis, assumptions and limitations.

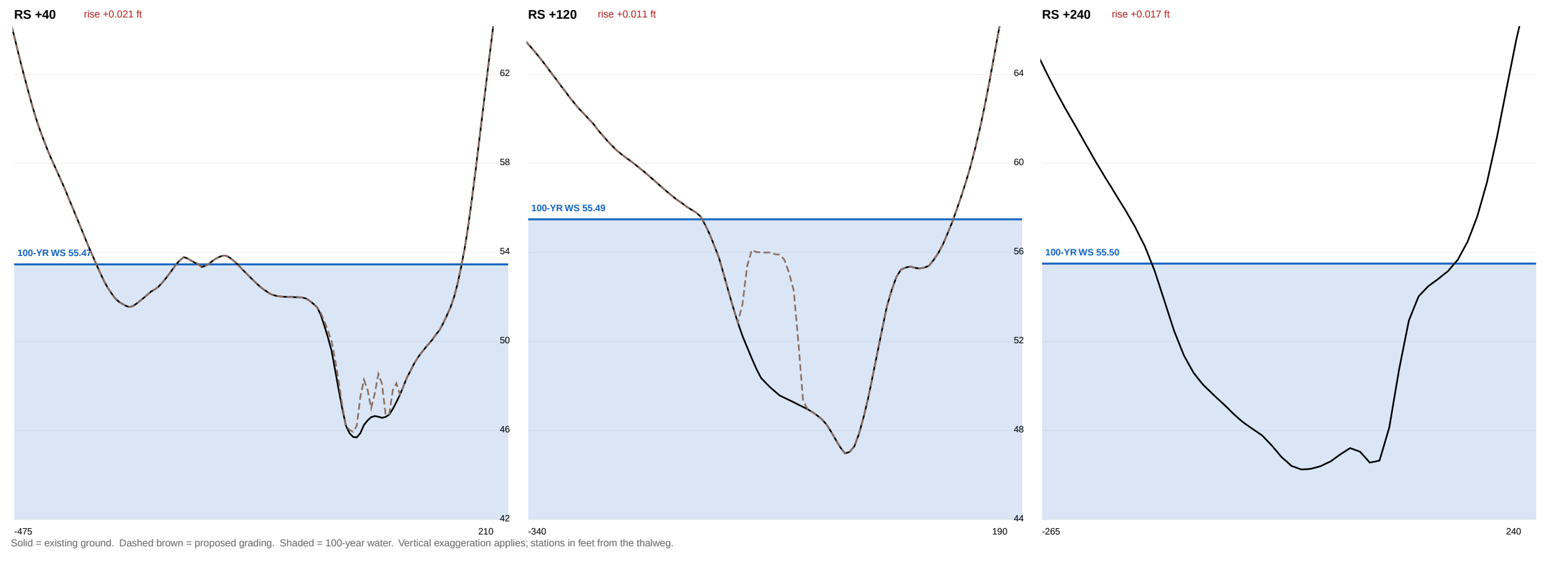
FLOODPLAIN DELINEATION — EXISTING AND PROPOSED, 100-YEAR



WATER-SURFACE PROFILE ALONG NORTH BRANCH BROAD CREEK



MODELLED CROSS SECTIONS — EXISTING AND PROPOSED



DESIGN DISCHARGES — NRCSTR-55, TYPE II					
STORM	P24 (IN)	Q (IN)	gw	PEAK (CFS)	
10-year	4.81	1.75	324	351	
25-year	6.01	2.63	350	571	
50-year	7.08	3.47	367	792	
100-year	8.29	4.48	382	1,061	
Area	397 ac	CN	68.0	Tc 0.86 hr	
CULVERT AT Q100					
36 in					
48 in					
60 in					
72 in					
84 in					
96 in					

EXISTING vs PROPOSED WATER SURFACE — 100-YEAR					
SECTION	EXISTING	PROPOSED	RISE (FT)		
RS +15	55.47	55.47	+0.000		
RS +40	55.47	55.49	+0.021	RISE	
RS +175	55.46	55.49	+0.030	RISE	
RS +120	55.49	55.50	+0.011	RISE	
RS +175	55.50	55.51	+0.016	RISE	
RS +240	55.50	55.52	+0.017	RISE	
RS +310	55.50	55.52	+0.018	RISE	
RS +390	55.50	55.52	+0.017	RISE	
RS +470	55.51	55.52	+0.017	RISE	
MAXIMUM			+0.030	NOT NO-RISE	
Storage removed below the 100-yr surface					
Total fill in the modelled reach					
Compensatory storage					

FLOODPLAIN ON THE LOTS, AND FREEBOARD — 100-YEAR				
LOT	AREA (SF)	FLOODED EXIST	FLOODED PROP	FILL < FLOOD
LOT 53	12,500	11,100 (89%)	5,900 (47%)	771 cy
LOT 54	23,500	23,500 (100%)	22,100 (94%)	2,516 cy
LOT 55	25,600	24,500 (96%)	23,000 (90%)	1,679 cy
LOT 56	13,900	2,800 (20%)	700 (5%)	113 cy
LOWEST FLOOR CHECK AGAINST WS EL 56.52				
LOT	FIN FLOOR	FREEBOARD	BASEMENT	FOUNDATION
LOT 53	59.24	+3.72	none	slab
LOT 54	57.57	+2.55	none	slab
LOT 55	57.60	+2.08	none	slab
LOT 56	61.02	+5.50	none	slab

BASIS, METHOD AND LIMITATIONS

HYDROLOGY NRCSTR-55 graphical peak discharge, Type II distribution. Drainage area delineated by D8 routing on the USGS 3DEP bare-earth DEM. Curve number composited from the county 2023 impervious surface and tree canopy layers and the NRCS soil survey. Rainfall from NOAA Atlas 14 Vol. 2 Ver. 3, partial duration series, at 38.7611 N 77.0115 W.

HYDRAULICS One-dimensional steady gradually-varied flow, standard step, conveyance subdivided at the bank stations, average-conveyance friction slope. Crossing analyzed to FHWA HDS-5 with both controls computed and the roadway treated as a broad-crested weir.

TERRAIN POGas Contour 2 F (2023), NAVD 88, interpolated to a 10 ft grid. THIS IS NOT A FIELD SURVEY.

DATUM NAVD 88 + NGVD 29 - 0.86 ft here, from NGS marks HV4762, HV4763 and HV4729. On that basis the 57 ft WSSC of FPS-770017 is about EL 56.3 NAVD 88, against EL 55.47 computed. To be confirmed by leveling to benchmarks H21A, H22A and H22B on the field topographic survey.

NOT DONE No field survey. The existing culvert has never been measured and its size, inverts and entrance are assumed. HEC-RAS was not run and no HEC-RAS files exist. FPS 200546, the controlling study of record, has not been obtained. Upstream stormwater facilities were not inventoried. No floodway was delineated and none exists on current mapping. No unredeemed or storage routing.

FLOODPLAIN OF RECORD FEMA FIRM panel 2403C02202E eff. 2016-09-16 maps these lots Zone X, outside the special flood hazard area, with no BFE and no floodway. That is a statement about FEMA mapping, not about flood risk. The county regulates this floodplain through the FPS series regardless.

EASEMENT The only patented floodplain easement in Indian Queen East (Pht 118-083, recorded 17 January 1984, 5.32 ac) lies entirely south of Fort Foote Road and covers none of Lots 53-56.

KEALEE

Design, build, deliver

PROJECT
Indian Queen East — Lots 53, 54, 55 and 56
ADDRESS
9588 · 9584 · 9580 · 9576 Fort Foote Road
JURISDICTION
Prince George's County, Maryland
ZONE
RSF-95

PLAT OF RECORD
Plat Book WWW 65, folio 60

SHEET
FP-101 — FLOODPLAIN STUDY AND DELINEATION, EXISTING AND PROPOSED

DISCIPLINE
Maryland Professional Engineer

HORIZONTAL DATUM
NAD 83, Maryland State Plane, US survey feet (EPSG:2248)

VERTICAL DATUM
NAVD 88

DATE
2026-09-11

STATUS
PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. No field survey. Culvert never measured. HEC-RAS not run. FPS 200546, the controlling study of record, not obtained. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. _____ Date _____

KEY FINDINGS

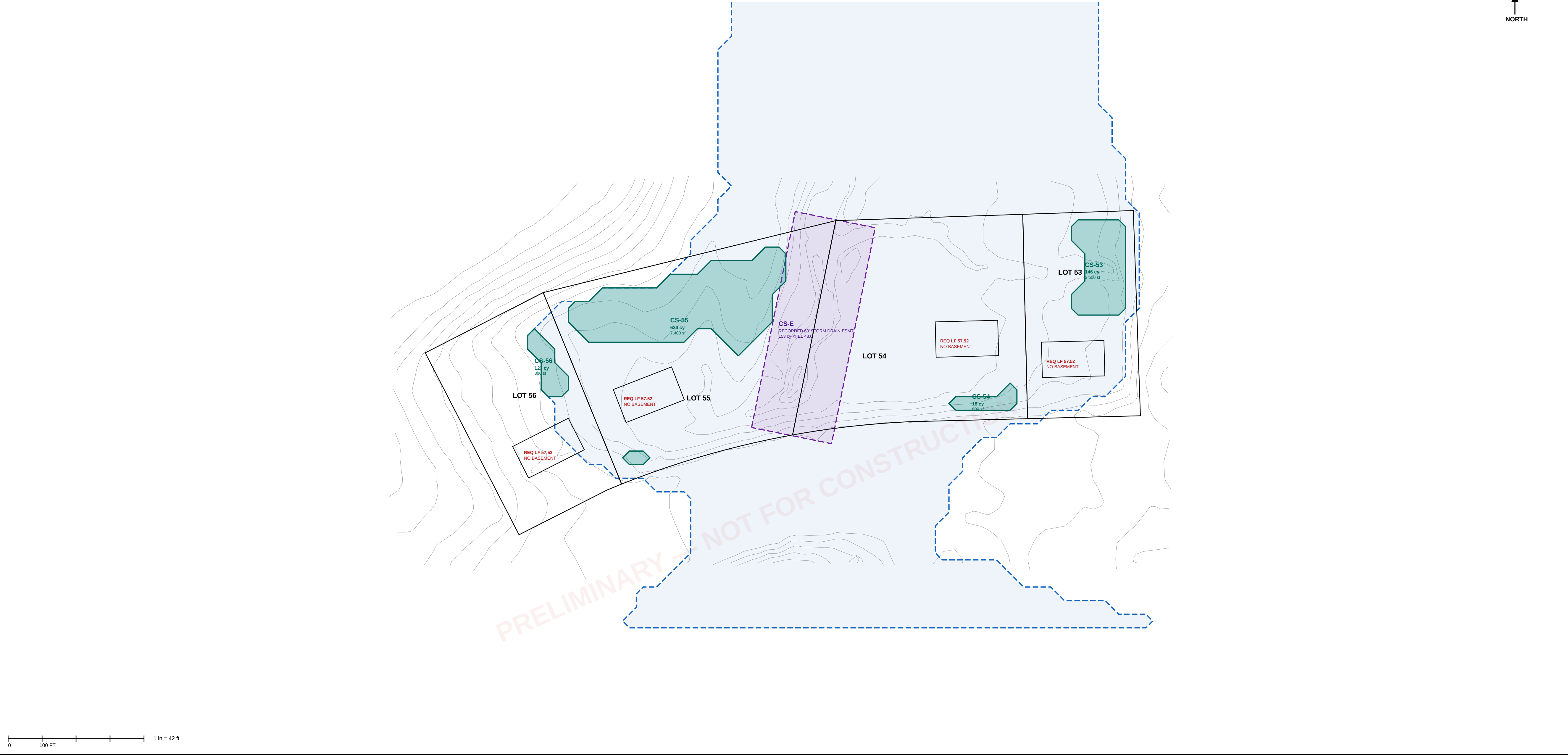
Contributing drainage area 397 ac (0.620 sq mi)
Composite curve number 68.0
Time of concentration 0.86 hr
Q100 at the crossing 1,061 cfs
Fort Foote Road sag EL 54.00
100-yr water surface, existing EL 55.47
100-yr water surface, proposed EL 55.52
Max rise, proposed vs existing +0.03 ft — NOT no-rise
Flood storage removed by fill 5,165 cy
Road overtops at Q100 956 cfs over the pavement

LEGEND

100-yr floodplain limit, EXISTING
100-yr floodplain limit, PROPOSED
Modelled cross section
Lot boundary of record
Existing contour (2 ft, county LIDAR)

COMPENSATORY STORAGE DESIGN — PLAN

Cells CS-53 to CS-56 excavated to EL 50.50; cell CS-E within the recorded 60 ft storm drain easement



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PROJECT

Indian Queen East — Lots 53, 54, 55 and 56

ADDRESS

9588 · 9584 · 9580 · 9576 Fort Foote Road

JURISDICTION

Prince George's County, Maryland

SHEET

FP-102 — MITIGATION ANALYSIS AND COMPENSATORY STORAGE DESIGN

DISCIPLINE

Maryland Professional Engineer

VERTICAL DATUM

NAVD 88

DATE

2026-09-11

STATUS

PRELIMINARY FEASIBILITY. NOT A SEALED FLOODPLAIN STUDY. Quantities are for concept evaluation and are not construction quantities. No professional engineer has reviewed or sealed this sheet.

PROFESSIONAL CERTIFICATION

Maryland P.E. No. _____ Date _____

DETERMINATIONS

RECOMMENDED: develop 53 and 56

Removal of all lots from floodplain

Controlling talwater, lower bound

Lowest ground, Lot 55

Basements

Required lowest floor

Compensation required

Compensation available on site

Crossing enlargement

Lot 54 flooded, 2-year event

Road overtopped

Storage to dry the rear yards

Upstream diversion benefit

53 + 56 only; compensation needed

available on 54 + 55 at EL 50

DEDICATE 54 AND 55

NOT ACHIEVABLE

EL 44.45

EL 44.66

EL 51.52

5,165 cy

1,070 cy

NOT RECOMMENDED

89%

BETWEEN 5- AND 10-YEAR

136,570 cy — NOT FEASIBLE

0.07 ft

891 cy

1,399 cy

DESIGN CRITERIA — COMPENSATORY STORAGE

1. Excavation floor EL 50.50, above the invert of the receiving channel, so each cell drains by gravity.

2. Cells located within the existing condition 100-year floodplain limit, being the area to be placed under floodplain easement.

3. Side slopes no steeper than 3:1, stabilized and revegetated, no retaining structure.

4. Minimum 10 ft from every property line and 20 ft from every structure.

5. Positive hydraulic connection to the recorded storm drain easement corridor at the cell invert.

6. Volume provided at the same elevation increments from which it is taken; stage-by-stage comparison required at final design.

RESERVOIR ROUTING — EXISTING CROSSING

STORM	INFLOW	OUTFLOW	ATTENUATED	PEAK STAGE
10-year	388 cfs	215 cfs	44.6%	EL 54.31
25-year	606 cfs	512 cfs	15.5%	EL 54.82
50-year	814 cfs	758 cfs	6.8%	EL 55.13
100-year	1,058 cfs	1,032 cfs	2.4%	EL 55.43

CROSSING ALTERNATIVES — DOWNSTREAM TRANSFER

OPTION	10-YR OUT	CHANGE	100-YR OUT	CHANGE
existing 36 in	215	+0	1,032	+0
48 in RCP	179	-36	1,023	-9
72 in RCP	275	+60	981	-52
twin 72 in RCP	333	+118	870	-162
8 x 6 ft box	307	+91	853	-80
10 x 8 ft box	347	+132	869	-163
twin 10 x 8 ft box	388	+173	974	-58
20 x 8 ft box (bridge-class)	388	+173	974	-58
twin 20 x 8 ft box	388	+173	1,058	+26

COMPENSATORY STORAGE SCHEDULE

CELL	LOT	AREA (SF)	MEAN DEPTH	VOLUME
CS-53	53	2,500	1.57 ft	146 cy
CS-54	54	600	0.79 ft	18 cy
CS-55	55	7,400	2.30 ft	630 cy
CS-56	56	800	4.16 ft	123 cy
CS-E	54/55	9,721	10 EL 48.0	153 cy

PROVIDED

REQUIRED by proposed grading

DEFICIT

FILL BELOW EL 55.53 — DISTRIBUTION

beneath dwelling footprints

within 10 ft of a dwelling

driveways, aprons, set-out

RESOLUTION: reduce fill. Vented stem-wall or

pier foundations and driveways at existing grade

bring the balance within the volume provided.

FLOODPLAIN REMOVAL AND FOUNDATION DETERMINATION

LOT	LOWEST GROUND	WS MUST FALL BELOW	REQUIRED DROP	ACHIEVABLE
LOT 53	EL 48.78	EL 48.78	6.69 ft	partial
LOT 54	EL 44.97	EL 44.97	10.50 ft	partial
LOT 55	EL 44.66	EL 44.66	10.81 ft	partial
LOT 56	EL 53.34	EL 53.34	2.13 ft	partial

Controlling talwater lower bound: EL 44.45

The upstream water surface cannot be lowered

below the downstream water surface.

LOWEST FLOOR DETERMINATION

LOT	PROPOSED FF	BASEMENT	REQ LOWEST FLOOR	STATUS
LOT 53	59.24	none	57.52	COMPLIES
LOT 54	57.57	none	57.52	COMPLIES
LOT 55	57.60	none	57.52	COMPLIES
LOT 56	61.02	none	57.52	COMPLIES

Basements are not permissible on any lot.

FREQUENCY OF INUNDATION — EXISTING GROUND

STORM	STAGE	LOT 54	LOT 55	ROAD
2-yr	EL 50.17	89%	41%	clear
5-yr	EL 53.02	97%	78%	clear
20-yr	EL 54.33	99%	92%	OVERTOPPED
25-yr	EL 54.82	100%	93%	OVERTOPPED
50-yr	EL 55.13	100%	95%	OVERTOPPED
100-yr	EL 55.43	100%	95%	OVERTOPPED

RECOMMENDED: DEVELOP LOTS 53 AND 56;

DEDICATE LOTS 54 AND 55 AS FLOODPLAIN

compensation needed by 53 + 56

available on 54 + 55 to EL 50.0

ratio provided to required

891 cy

1,399 cy

1.6 : 1

KEALEE

Design, build, deliver

PROJECT
Indian Queen East — LOT 53 & LOT 54 & LOT 55 & LOT 56
ADDRESS
9588 Fort Foote Rd · 9584 Fort Foote Rd · 9580 Fort Foote Rd · 9576 Fort Foote Rd
JURISDICTION
Prince George's County, Maryland
ZONE
RSF-95
SHEET
C-001 — Site Development Plan — Existing, Proposed, Grading, Paving and Landscape
SCALE
1" = 30'
STATUS
PRELIMINARY
SHEET NO.
1 OF 3
COORDINATES
EPSG 2248
H: NAD83
V: NAVD88 (feet)

PLAN TYPE AND PREPARATION

PLAN TYPE SITE DEVELOPMENT / FINE GRADING
SHEET C-001 — Site Development Plan — Existing, Proposed, Grading, Paving and Landscape
DISCIPLINE Maryland Professional Engineer / Surveyor (see divided responsibility)
DRAWN BY KEALEE SITE-PLAN ENGINEER
DESIGNED BY
CHECKED BY

REVISIONS

NO.	DATE	DESCRIPTION

INDEX OF DRAWINGS

- 1 C-001 Site Development Plan — Existing, Proposed, Grading, Paving and
2 L-100 Landscape and Tree Canopy Plan
3 C-400 Grading and Drainage Plan

SITE DATA

ZONE	RSF-95
TAX / PARCEL ID	—
GROSS TRACT AREA	75,178 SF
LOT 53 AREA	12,389 SF
LOT 54 AREA	23,371 SF
LOT 55 AREA	25,733 SF
LOT 56 AREA	13,690 SF
NET LOT AREA	75,184 SF
LOT 53 FRONTAGE	50' MIN
LOT 54 FRONTAGE	50' MIN
LOT 55 FRONTAGE	50' MIN
LOT 56 FRONTAGE	50' MIN
SETBACKS	REQUIRED PROVIDED
FRONT YARD	30' 30'
SIDE YARD	8' 8'
REAR YARD	20' 20'
COVERAGE	MAX PROPOSED
LOT COVERAGE	30% 6.4%
BUILDING FOOTPRINT	— 4,784 SF
DATUM	
HORIZONTAL	EPSG 2248 NAD83
VERTICAL	NAVD88 (feet)

SITE ANALYSIS

1. Gross tract area 75,178 SF
2. Area of dwelling 4,784 SF
3. Wooded area NOT ESTABLISHED
4. Floodplain area NOT ESTABLISHED
5. Net lot area 75,184 SF
6. TOTAL AREA DISTURBED 41,662 SF (AT LEAST)
7. SWM REQUIRED — Sec 32-174(a)(3) YES — DISTURBANCE EXCEEDS 5,000 SF

BUILDING DATA

	G	B	FF	SF	HT	STY
9588 Fort Foote Rd	58.91	—	59.24	58.24	20'	2
9584 Fort Foote Rd	57.24	—	57.57	56.57	20'	2
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9576 Fort Foote Rd	60.69	—	61.02	60.02	20'	2

G: garage slab · B: basement · FF: finished floor · SF: subfloor · HT: height · STY: stories · A DASH IS NOT ZERO: the elevation has not been established and must be set from a field-run topographic survey before construction.

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CATCHMENT	AREA SF	IMPERV.	PRE Q	POST Q	INCREASE	WQv CF
LOT 53	12,389	15.1%	0.49	0.69	0.20	192
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LOT 55	25,733	8.2%	1.01	1.24	0.23	266
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PROJECT TOTAL	75,184	10.8%	2.96	3.84	0.88	924

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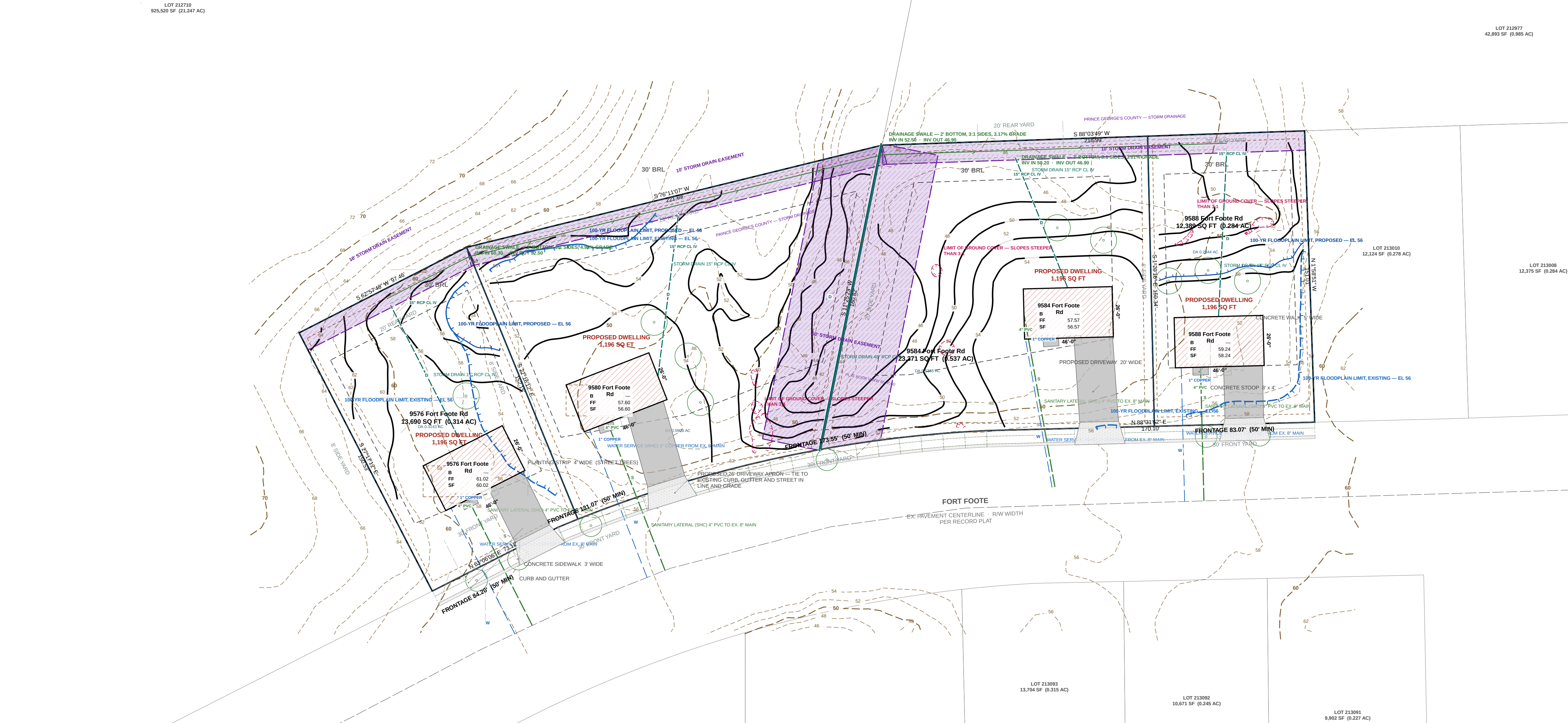
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I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE	DATE
LICENSE NO.	EXPIRATION DATE
SEAL	
REVISIONS	
— NONE —	



SOILS TABLE

USDA NRCS SSURGO — PGC Code Sec. 32-133(a)(13)

MAP UNIT	MAP UNIT NAME	K-FACT	HYRIC	HYD. GRP	DRAINAGE CLASS
AaB	Adelphia silt loam, 2 to 5 percent slopes complex	37	No	C	Moderately well drained
ACA	Adelphia-Aquasco complex, 0 to 2 percent slopes	43	No	D	Somewhat poorly drained
ADA	Adelphia-Holmdel complex, 0 to 2 percent slopes	24	No	B/D	Somewhat poorly drained
ADB	Adelphia-Holmdel complex, 2 to 5 percent slopes	24	No	B/D	Somewhat poorly drained
ADC	Adelphia-Holmdel complex, 5 to 10 percent slopes	24	No	B/D	Somewhat poorly drained

DRAINAGE SWALE SCHEDULE

Rational method, NOAA Atlas 14 · 100%: 0.0131 0.041, 2' bottom, 3:1 sides

REACH	LENGTH	GRADE	D.A. (AC)	Q10 (CFS)	Q100 (CFS)	DEPTH	V100 (FPS)	INV IN	INV OUT	CUT
ST-1 — ST-2	110'	3.01%	0.2844	0.59	0.78	0.18'	1.73	50.20	46.90	1.0'
ST-4 — ST-3	157'	4.98%	0.3143	0.65	0.86	0.16'	2.12	60.30	52.50	1.0'
ST-3 — ST-2	177'	3.17%	0.9051	1.87	2.47	0.33'	2.51	52.50	46.90	1.0'

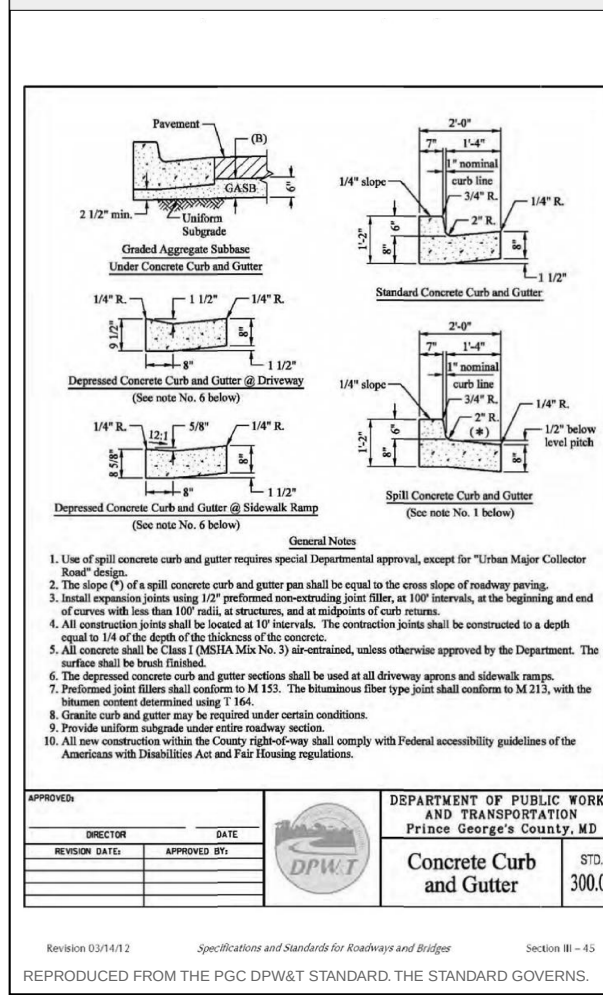
GRADE CONTROL SCHEDULE

POINT	LOT	DESCRIPTION	GROUND	SWALE INV
ST-1	LOT 53	Swale grade break / invert control	51.20	50.20
ST-2	LOT 54	Swale grade break / invert control	47.90	46.90
ST-3	LOT 55	Swale grade break / invert control	53.50	52.50
ST-4	LOT 56	Swale grade break / invert control	61.30	60.30
OUT	—	Swale grade break / invert control	48.90	—

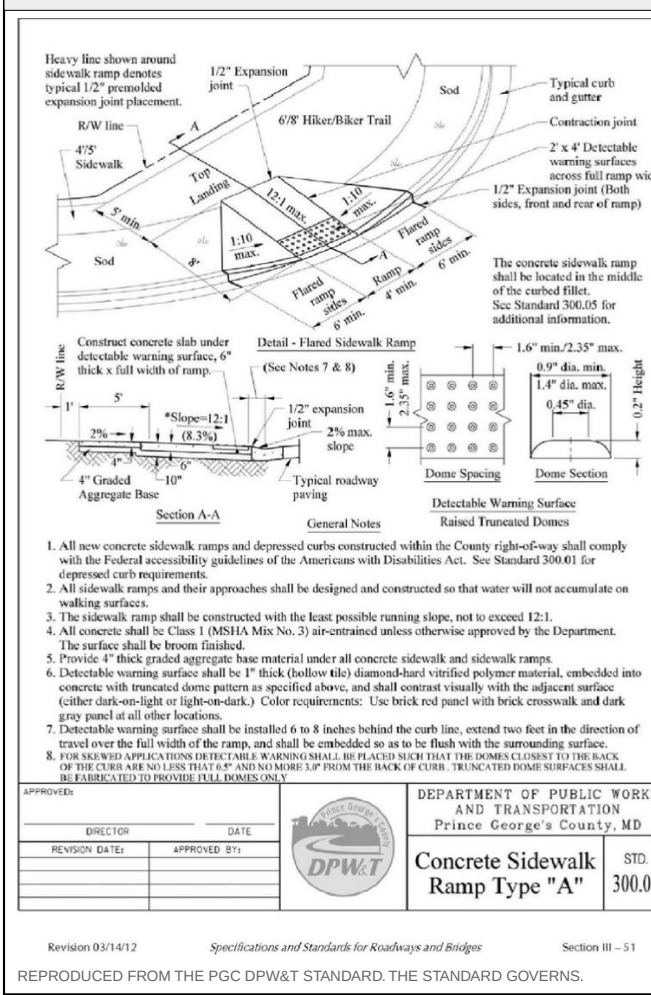
SYSTEM AT THE LOW POINT: 1.7280 AC, Q10 3.57 CFS, Q100 4.71 CFS.

MAX VELOCITY 2.61 FPS AT Q100 — WITHIN THE 5 FPS PERMISSIBLE FOR ESTABLISHED GRASS, 0.6 FT FREEBOARD THROUGHOUT. THE SWALE DISCHARGES TO THE RECORDED 5485 STORM DRAIN EASEMENT, ALONGSIDE THE 40' ROP CL IV EXTENDING THE FORT FOOTE ROAD CULVERT INLET. THAT MAIN IS BURIED, OD 4.67 FT, 1.5 FT MIN COVER OVER THE CROWN — FINISHED GRADE OVER IT SHALL BE NO LOWER THAN INVERT + 6.17 FT. FILL WHERE EXISTING GROUND IS LOWER. SWALE GRADES AND INVERTS TO BE SET FROM THE FIELD-RUN TOPOGRAPHIC SURVEY. ESTABLISH SOD BEFORE THE CONTRIBUTING AREA IS STABILIZED.

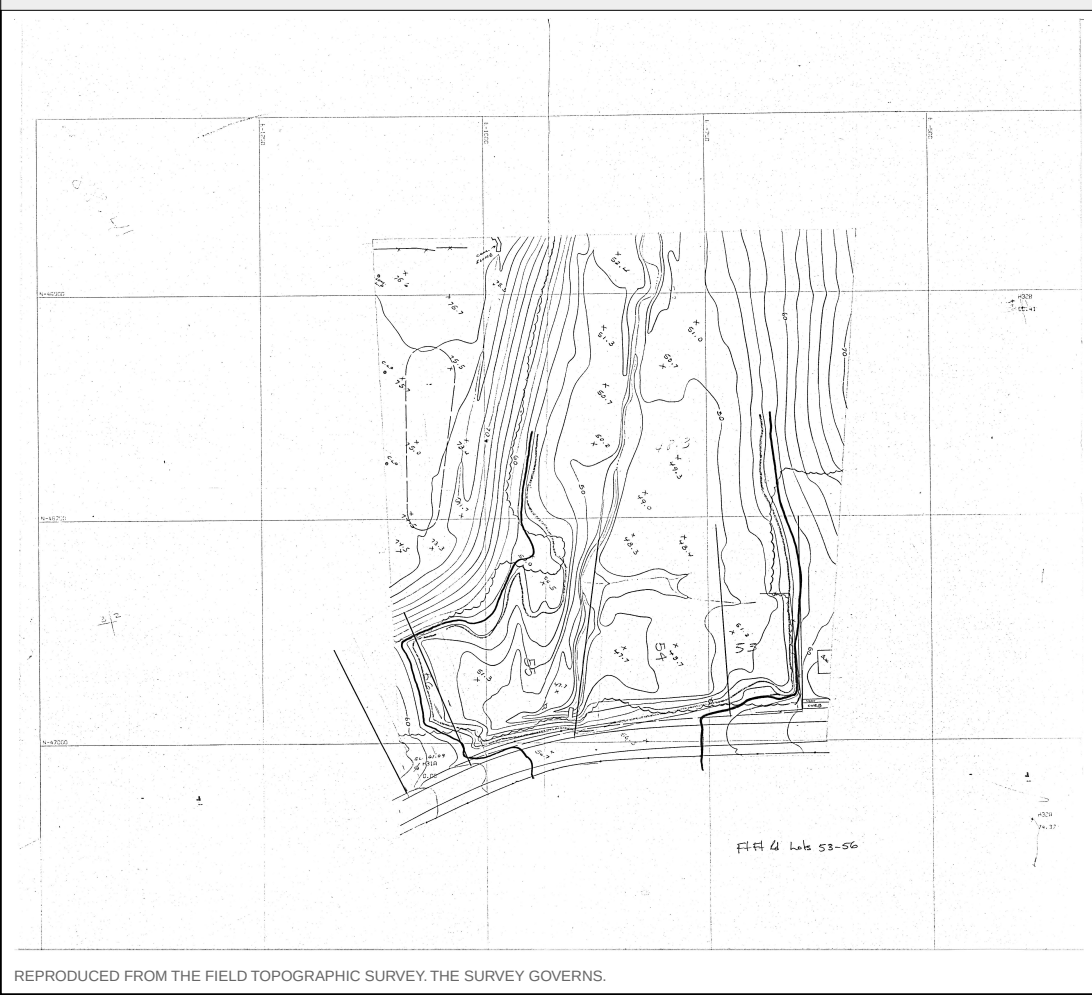
CONCRETE CURB AND GUTTER — PGC DPW&T STD. 300.01



CONCRETE SIDEWALK RAMP TYPE A — PGC DPW&T STD. 300.07



FIELD TOPOGRAPHIC SURVEY — EXISTING CONTOURS — SURVEYED — BM H31A EL. 61.09



GRAPHIC SCALE — 1" = 30'
0 30 60 90 120 FEET

KEALEE

Design, build, deliver

PROJECT
Indian Queen East — LOT 53 & LOT 54 & LOT 55 & LOT 56
ADDRESS
9588 Fort Foote Rd · 9584 Fort Foote Rd · 9580 Fort Foote Rd · 9576 Fort Foote Rd
JURISDICTION
Prince George's County, Maryland
ZONE
RSF-95
SHEET
L-100 — Landscape and Tree Canopy Plan
SCALE
1" = 30'
STATUS
PRELIMINARY
SHEET NO.
2 OF 3
COORDINATES
EPSG:2248
H: NAD83
V: NAVD88 (feet)

PLAN TYPE AND PREPARATION

PLAN TYPE SITE DEVELOPMENT / FINE GRADING
SHEET L-100 — Landscape and Tree Canopy Plan
DISCIPLINE Landscape Architect / Qualified Professional
DRAWN BY KEALEE SITE-PLAN ENGINE
DESIGNED BY
CHECKED BY

REVISIONS

NO.	DATE	DESCRIPTION

INDEX OF DRAWINGS

- C-001 Site Development Plan — Existing, Proposed, Grading, Paving and
- L-100 Landscape and Tree Canopy Plan
- C-400 Grading and Drainage Plan

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LOT 53 AREA	12,389 SF
LOT 54 AREA	23,371 SF
LOT 55 AREA	25,733 SF
LOT 56 AREA	13,690 SF
NET LOT AREA	75,184 SF
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LOT 54 FRONTAGE	50' MIN
LOT 55 FRONTAGE	50' MIN
LOT 56 FRONTAGE	50' MIN
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FRONT YARD	30'
SIDE YARD	8'
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COVERAGE MAX	PROPOSED
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BUILDING FOOTPRINT	—
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SITE ANALYSIS

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SIGNATURE DATE

LICENSE NO. EXPIRATION DATE

REVISIONS

— NONE —

TREE CANOPY SCHEDULE

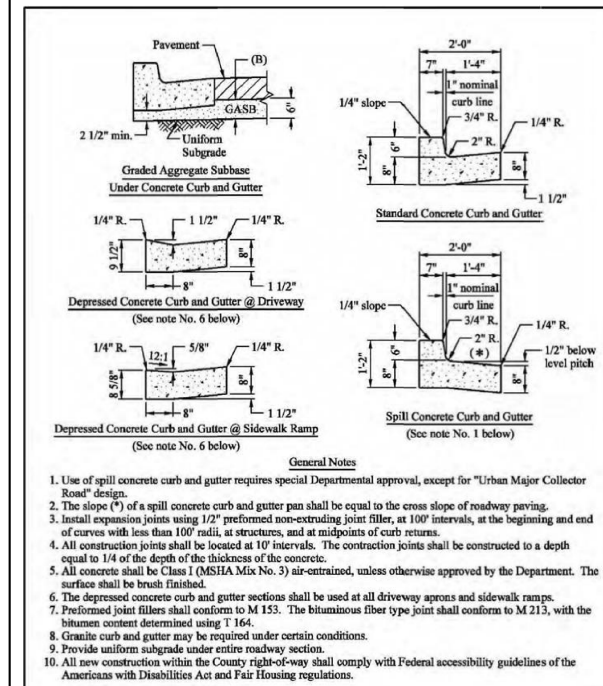
Sec. 25-128 Table 1 as amended by CB-021-2024 — 20% of NET tract area on RSF-95, met within 8'. Per-lot planting per Landscape Manual Sec. 4.1(c)(3). Specif trees in the right-of-way count toward canopy under Sec. 25-129.

LOT	NET AREA SF	CANOPY REQ 20% SF	SHADE PROV / REQ	ORN-EVON PROV / REQ	STREET TREES	10-YR CANOPY PROVIDED
LOT 53	12,389	2,478	3 / 3	0 / 2 SHORT	2	NOT ESTABLISHED
LOT 54	23,371	4,674	2 / 4 SHORT	0 / 3 SHORT	1	NOT ESTABLISHED
LOT 55	25,733	5,147	3 / 4 SHORT	0 / 3 SHORT	1	NOT ESTABLISHED
LOT 56	13,690	2,738	1 / 3 SHORT	0 / 2 SHORT	2	NOT ESTABLISHED
TOTAL	75,184	15,037	9 / 14	0 / 10	6	—

TEN-YEAR CANOPY IS CREDITED FOR SPECIES FROM THE PRINCE GEORGE'S COUNTY LANDSCAPE MANUAL, NO SPECIES HAS BEEN SELECTED ON THIS SET, SO THE CANOPY PROVIDED IS NOT ESTABLISHED AND THE SCHEDULE IS NOT YET BALANCED. EXISTING TREES PRESERVED ALSO COUNT. NO TREE IS PLANTED WITHIN A STORM DRAIN EASEMENT; THE 60 FT 5405 EASEMENT CARRIES THE 48 IN RCP MAIN AND THE 10 FT REAR EASEMENT CARRIES THE DRAINAGE SWALE, BOTH ARE SHOWN ON THIS SHEET AND BOTH ARE EXCLUDED FROM PLANTABLE AREA.

A MAJOR OF THE CANOPY REQUIREMENT IS AVAILABLE UNDER SEC. 25-130.
PER-LOT PLANTING IS SHOWN ON EVERY LOT AS DRAWN, LANDSCAPE MANUAL, SEC. 4.1(c)(3) REQUIRES A MAJOR SHADE AND 3 ORNAMENTAL, OR EVERGREEN TREES ON A LOT OF 20,000 SF OR MORE, AND 3 AND 3 ON A LOT OF 5,000 TO 19,999 SF. NO ORNAMENTAL, OR EVERGREEN TREES ARE DRAWN, AT LEAST ONE SHADE TREE MUST STAND ON THE SOUTH OR WEST SIDE WITHIN 30 FT OF THE DWELLING, AND AT LEAST ONE REQUIRED TREE IN THE FRONT YARD. SPECIES AND FINAL LOCATIONS BY THE LANDSCAPE ARCHITECT.
THE CANOPY REQUIREMENT MUST BE MET ON-SITE UNLESS A VARIANCE IS APPROVED, AND WHERE A GRADING PERMIT IS THE ONLY APPLICATION THE CANOPY NOTES BELONG ON THE GRADING PLAN — ENVIRONMENTAL TECHNICAL MANUAL PART D, SEC. 3.0.

CONCRETE CURB AND GUTTER — PGC DPW&T STD. 300.01



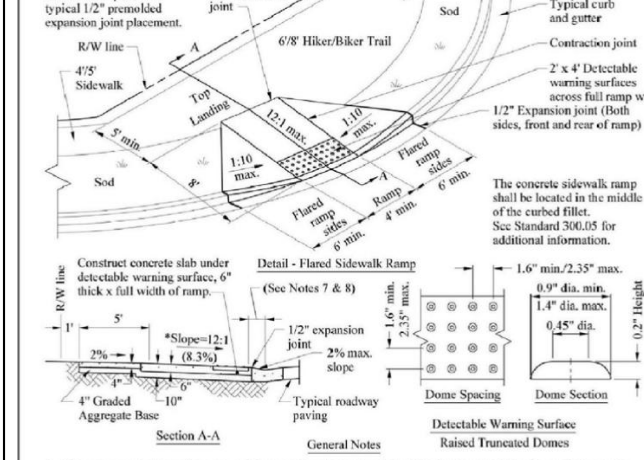
APPROVED	DATE	DESIGNED BY	DATE

Version 03/04/12
Specification and Standards for Roadways and Bridges
Section B-11
REPRODUCED FROM THE PGC DPW&T STANDARD, THE STANDARD GOVERNS.

GRAPHIC SCALE — 1" = 30'

0 30 60 90 120 FEET

CONCRETE SIDEWALK RAMP TYPE A — PGC DPW&T STD. 300.07



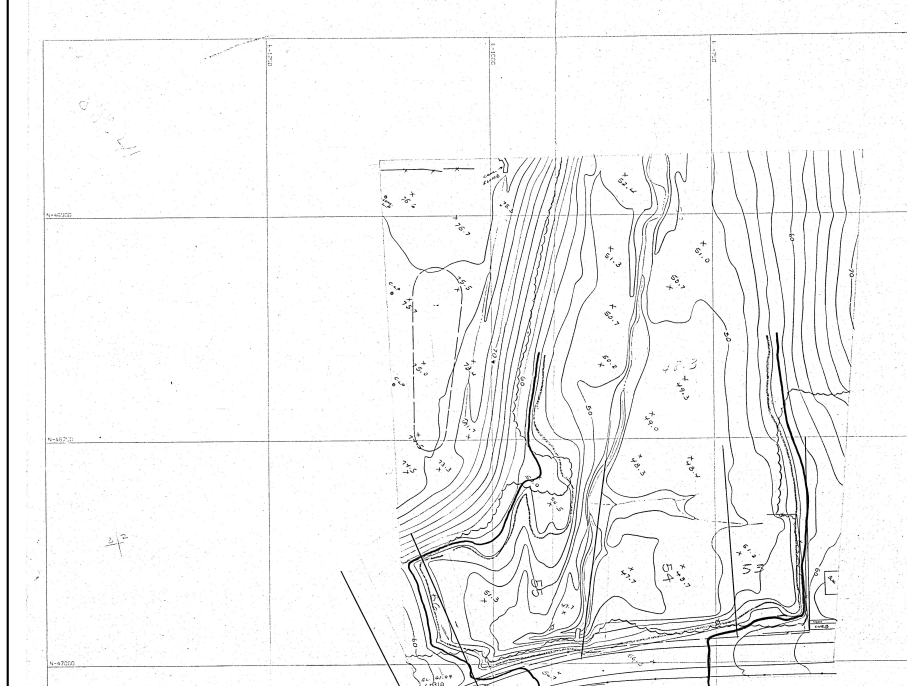
- All new concrete sidewalk ramps and depressed curbs constructed within the County right-of-way shall comply with the Federal accessibility guidelines of the Americans with Disabilities Act. See Standard 300.01 for expanded curb requirements.
- All sidewalk ramps and curb requirements shall be designed and constructed so that water will not accumulate on the sidewalk.
- The sidewalk ramp shall be constructed with the base parallel to the curb, and a crown 1/2" to 1" wide.
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FIELD TOPOGRAPHIC SURVEY — EXISTING CONTOURS — SURVEYED — BM H31A EL. 61.09



APPROVED	DATE	DESIGNED BY	DATE

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KEALEE

Design, build, deliver

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JURISDICTION
Prince George's County, Maryland
ZONE
RSF-95
SHEET
C-400 — Grading and Drainage Plan
SCALE
1" = 30'
STATUS
PRELIMINARY
SHEET NO.
3 OF 3
COORDINATES
EPSG 2248
H: NAD83
V: NAVD88 (feet)

PLAN TYPE AND PREPARATION

PLAN TYPE SITE DEVELOPMENT / FINE GRADING
SHEET C-400 — Grading and Drainage Plan
DISCIPLINE Maryland Professional Engineer
DRAWN BY KEALEE SITE-PLAN ENGINE
DESIGNED BY
CHECKED BY

REVISIONS

NO.	DATE	DESCRIPTION

INDEX OF DRAWINGS

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7. NO DRIVEWAY CULVERT IS REQUIRED. THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION. NOT AN OPEN DITCH SECTION. GUTTER FLOW IS CARRIED THROUGH EACH ENTRANCE BY THE DEPRESSED CURB AT THE APRON PER DPW&T STD. 300.01. NOTE 6. ON-LOT DRAINAGE IS COLLECTED BY THE REAR-YARD SWALE IN THE REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE STATE OF MARYLAND.

SIGNATURE	DATE
LICENSE NO.	EXPIRATION DATE
SEAL	

IF REQUIRED NOTE(S) DO NOT FIT THIS SHEET, GRADING CERTIFICATE, STABILIZATION COMPLIANCE PREAMBLE, STANDARD STABILIZATION NOTE, THEY ARE REQUIRED AND ARE NOT ON ANY SHEET IN THIS SET.

LOT 212710
925,520 SF (0.147 AC)

LOT 212977
42,893 SF (0.985 AC)

LOT 213008
12,378 SF (0.284 AC)

LOT 213093
13,704 SF (0.315 AC)

LOT 213092
10,671 SF (0.245 AC)

LOT 213091
9,902 SF (0.227 AC)

SOILS TABLE USDA NRCS SSURGO — PGC Code Sec. 32-133(a)(13)

MAP UNIT	MAP UNIT NAME	K-FACT	HYRIC	HYD. GRP	DRAINAGE CLASS
AaB	Adelphia silt loam, 2 to 5 percent slopes complex	37	No	C	Moderately well drained
ACA	Adelphia-Aquasco complex, 0 to 2 percent slopes	43	No	D	Somewhat poorly drained
ADA	Adelphia-Holmdel complex, 0 to 2 percent slopes	24	No	B/D	Somewhat poorly drained
ADB	Adelphia-Holmdel complex, 2 to 5 percent slopes	24	No	B/D	Somewhat poorly drained
ADC	Adelphia-Holmdel complex, 5 to 10 percent slopes	24	No	B/D	Somewhat poorly drained

DRAINAGE SWALE SCHEDULE Rational method, NOAA Atlas 14 · 100Y:0.023(0.041), 2' bottom, 3:1 sides

REACH	LENGTH	GRADE	D.A. (AC)	Q10 (CFS)	Q100 (CFS)	DEPTH	V100 (FPS)	INV IN	INV OUT	CUT
ST-1 — ST-2	110'	3.01%	0.2844	0.59	0.78	0.18'	1.73	50.20	46.90	1.0'
ST-4 — ST-3	157'	4.98%	0.3143	0.65	0.86	0.16'	2.12	60.30	52.50	1.0'
ST-3 — ST-2	177'	3.17%	0.9051	1.87	2.47	0.33'	2.51	52.50	46.90	1.0'

GRADE CONTROL SCHEDULE

POINT	LOT	DESCRIPTION	GROUND	SWALE INV
ST-1	LOT 53	Swale grade break / invert control	51.20	50.20
ST-2	LOT 54	Swale grade break / invert control	47.90	46.90
ST-3	LOT 55	Swale grade break / invert control	53.50	52.50
ST-4	LOT 56	Swale grade break / invert control	61.30	60.30
OUT	—	Swale grade break / invert control	48.90	—

SYSTEM AT THE LOW POINT: 1.7260 AC, Q10 3.57 CFS, Q100 4.71 CFS.
MAX VELOCITY 2.61 FPS AT Q100 — WITHIN THE 5 FPS PERMISSIBLE FOR ESTABLISHED GRASS, 0.6 FT FREEBOARD THROUGHOUT.
THE SWALE DISCHARGES TO THE RECORDED 5485 STORM DRAIN EASEMENT, ALONGSIDE THE 40' ROP CL IV EXTENDING THE FORT FOOTE ROAD CULVERT INLET. THAT MAIN IS BURIED, OD 4.67 FT, 1.5 FT MIN COVER OVER THE CROWN — FINISHED GRADE OVER IT SHALL BE NO LOWER THAN INVERT + 6.17 FT. FILL WHERE EXISTING GROUND IS LOWER. SWALE GRADES AND INVERTS TO BE SET FROM THE FIELD-RUN TOPOGRAPHIC SURVEY. ESTABLISH SOD BEFORE THE CONTRIBUTING AREA IS STABILIZED.

CONCRETE CURB AND GUTTER — PGC DPW&T STD. 300.01

Diagram showing cross-sections of concrete curb and gutter for various conditions: 1. Typical curb and gutter, 2. Curb and gutter with drainage, 3. Curb and gutter with drainage and sidewalk, 4. Curb and gutter with drainage and sidewalk and driveway apron. Includes notes on materials, dimensions, and construction details.

DEPARTMENT OF PUBLIC WORKS AND TRANSPORTATION
Prince George's County, MD
Concrete Curb and Gutter
300.01

CONCRETE SIDEWALK RAMP TYPE A — PGC DPW&T STD. 300.07

Diagram showing cross-sections of concrete sidewalk ramp type A for various conditions: 1. Typical sidewalk ramp, 2. Sidewalk ramp with drainage, 3. Sidewalk ramp with drainage and sidewalk, 4. Sidewalk ramp with drainage and sidewalk and driveway apron. Includes notes on materials, dimensions, and construction details.

DEPARTMENT OF PUBLIC WORKS AND TRANSPORTATION
Prince George's County, MD
Concrete Sidewalk Ramp Type "A"
300.07

FIELD TOPOGRAPHIC SURVEY — EXISTING CONTOURS — SURVEYED — BM H31A EL. 61.09

